NIFTY100 FINANCIAL INTELLIGENCE PLATFORM

Sprint 1 – Data Foundation Report

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1.Executive Summary

The objective of Sprint 1 was to establish a reliable data foundation for the Nifty100 Financial Intelligence Platform. This sprint focused on data ingestion, validation, normalization, database design, and data quality assurance.

The project involved processing multiple financial datasets containing information related to company fundamentals, profit and loss statements, balance sheets, cash flows, financial ratios, sectors, peer groups, stock prices, annual reports, and company insights.

A complete ETL (Extract, Transform, Load) pipeline was developed using Python and SQLite. Data quality validation rules were implemented to ensure data consistency, uniqueness, referential integrity, and completeness. The final outcome of Sprint 1 is a fully populated SQLite database that serves as the foundation for subsequent analytical modules.

2.Project Objectives

The primary objectives of Sprint 1 were:

• Create a structured project environment for data engineering activities.

• Design and implement an ETL pipeline for processing raw Excel datasets.

• Normalize and validate source data.

• Build a relational SQLite database.

• Implement data quality validation rules.

• Generate validation reports and audit files.

• Prepare the project foundation for advanced analytics and dashboard development.

3.Data Sources

Dataset	Description
companies	Company profile information, website links, ROE and ROCE
profitandloss	Revenue, expenses, operating profit, net profit, EPS, and dividend payout details
balancesheet	Assets, liabilities, reserves, borrowings, and equity information
cash Flow	Operating, investing, financing, and net cash flow activities
Analysis	Growth metrics, CAGR calculations, and historical performance indicators
documents	Annual report links and company document references
prosandcons	Qualitative strengths and weaknesses of companies
Financial_ratios	Profitability, leverage, liquidity, and efficiency ratios
Peer_groups	Industry peer comparison groups and benchmark companies
sectors	Sector classification, market-cap category, and index weight information
Stock_prices	Historical stock price data including OHLC, volume, and adjusted close values

These datasets were collected and integrated to provide a comprehensive view of Nifty100 companies. Together, they support financial analysis, data validation, company screening, sector comparison, and future analytical modules of the Financial Intelligence Platform.

4.Technology Stack

The Nifty100 Financial Intelligence Platform was developed using the following technologies and tools:

Category	**Technology**
Programming Language	Python
Data Processing	Pandas
Database	SQLite
Testing Framework	Pytest
Version Control	Git
Repository Hosting	GitHub
Development Environment	Visual Studio Code

These technologies were selected to support efficient data ingestion, validation, transformation, testing, and database management throughout Sprint 1 of the project.

5.ETL Pipeline Design

The ETL pipeline was designed to automate the movement of data from raw Excel files into the SQLite database.

The workflow consisted of:

Step 1: Data Extraction
Raw Excel datasets were loaded into Pandas DataFrames.

Step 2: Data Normalization
Company identifiers and year formats were standardized to ensure consistency.

Step 3: Data Validation
Primary key, foreign key, and uniqueness checks were performed.

Step 4: Database Loading
Validated datasets were loaded into SQLite tables.

Step 5: Quality Verification
Row counts and validation reports were generated.

The ETL architecture ensured repeatable and scalable data processing.

6.Database Design

A relational SQLite database named "nifty100.db" was created.

The database contains the following tables:

- companies
- profit and loss
- balance sheet
- cashflow
- analysis
- documents
- pros and cons
- financial_ratios
- peer_groups
- sectors
- stock_prices

The schema was designed to support future analytical modules including screening, scoring, peer comparison, and dashboard visualization.

7.Data Validation Framework

Several validation rules were implemented to ensure data quality.

DQ-01 Primary Key Validation

Verified uniqueness of primary key values.

DQ-02 Company-Year Uniqueness

Validated that each company-year combination appears only once.

DQ-03 Foreign Key Validation

Ensured valid company references across datasets.

DQ-04 Null Value Analysis

Measured missing values across all tables.

DQ-05 Negative Value Analysis

Evaluated negative financial values and assessed business validity.

DQ-06 Data Type Validation

Verified numeric, text, and date fields.

DQ-07 Range Validation

Confirmed stock prices remained within valid ranges.

DQ-08 Date Validation

Validated stock market date records.

Validation reports were generated and stored for auditing purposes.

8.Data Loading Results

The final database contains the following records:

Table	Records
companies	92
profitandloss	1276
Balance sheet	1312
cashflow	1187
analysis	20
documents	1585
Pros and cons	16
financial_ratios	1184
peer_groups	56
sectors	92
stock_prices	5520

All datasets were successfully loaded into the SQLite database.

9.Testing Results

Unit tests were executed for normalization and validation modules.

All implemented tests passed successfully with zero failures.

The successful execution of all tests confirms the reliability of the ETL pipeline, validation framework, and data processing logic.

10.Deliverables Generated

The following project deliverables were created:

• nifty100.db
• schema. sql
• validation_failures.csv
• load_audit.csv
• dq_report.csv
• final_validation_report.txt
• README.md

These files collectively document the project implementation and validation outcomes.

11.Project Folder Structure

The project was organized using a structured folder hierarchy to improve maintainability and scalability.

```
nifty100_capstone/
├── data/
│   └── raw/
├── db/
│   ├── nifty100.db
│   ├── schema.sql
│   └── create_database.py
├── src/
│   └── etl/
├── tests/
├── notebooks/
├── output/
├── README.md
└── requirements.txt
```

This structure ensured clear separation between source data, database assets, ETL scripts, testing modules, reports, and project documentation.

12.Key Achievements

The following milestones were successfully achieved during Sprint 1:

- Developed an end-to-end ETL pipeline using Python.
- Created a normalized SQLite database containing financial information for Nifty100 companies.
- Implemented multiple data quality validation rules.
- Generated audit and validation reports.
- Successfully loaded more than 11,000 records across multiple datasets.
- Achieved 100% pass rate in implemented unit tests.
- Maintained version control using Git and GitHub.
- Prepared the project foundation for future analytical modules.

13.Future Scope

The data foundation created during Sprint 1 will support future development phases of the Nifty100 Financial Intelligence Platform.

Planned future enhancements include:

- Financial Ratio Engine
- Investment Screening Module
- Financial Health Scoring System
- Peer Comparison Engine
- Sector Analytics
- Interactive Dashboards
- Report Generation
- Portfolio Tracking and Alerts

These modules will transform the project from a data repository into a comprehensive financial analytics platform.

14.Challenges Encountered

Several challenges were encountered during development:

• Duplicate company-year records in source datasets.

• Foreign key inconsistencies across datasets.

• Excel header variations requiring custom loading logic.

• Database schema mismatches during data loading.

These issues were identified and resolved through iterative validation and testing.

15.Appendix

The following supporting artifacts were generated as part of Sprint 1:

• SQLite Database (nifty100.db)

• Data Quality Report (dq_report.csv)

• Validation Failures Report (validation_failures.csv)

• Load Audit Report (load_audit.csv)

• Final Validation Report (final_validation_report.txt)

• SQL Schema Definition (schema.sql)

• Unit Test Suite

• GitHub Repository

These artifacts provide evidence of successful completion of Sprint 1 objectives and support future development activities.

16.Project Evidence

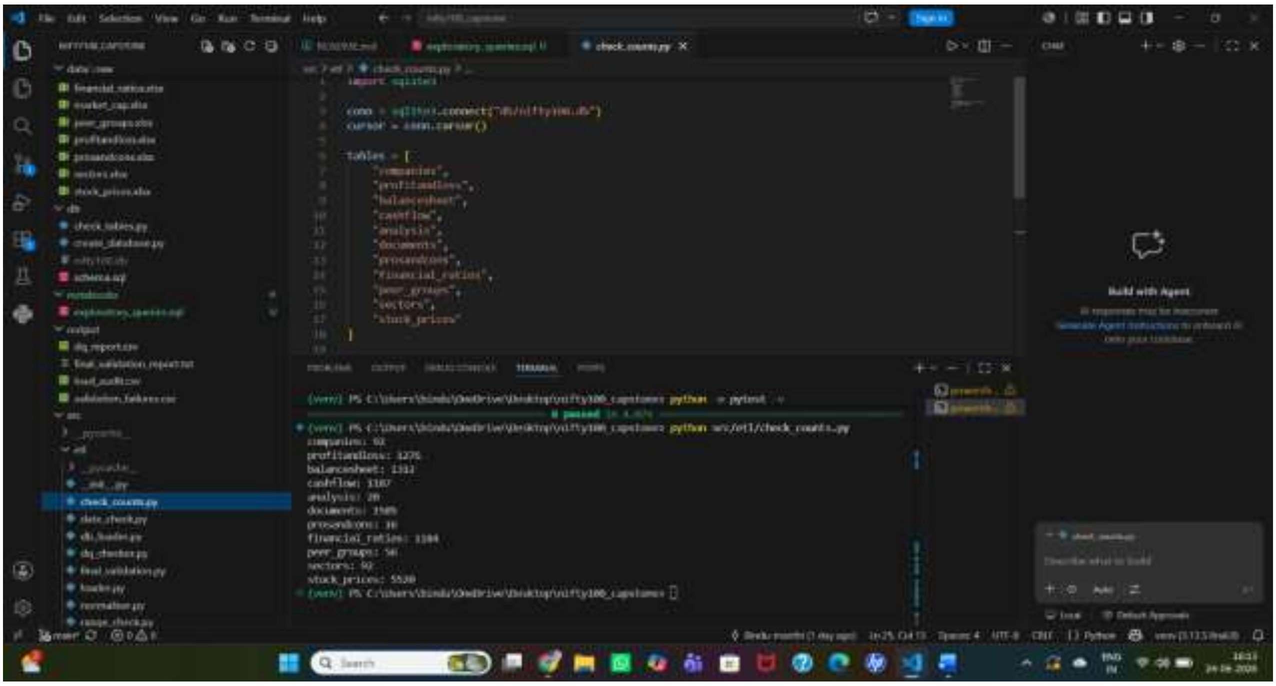

Figure 1: Final database row counts for all tables.

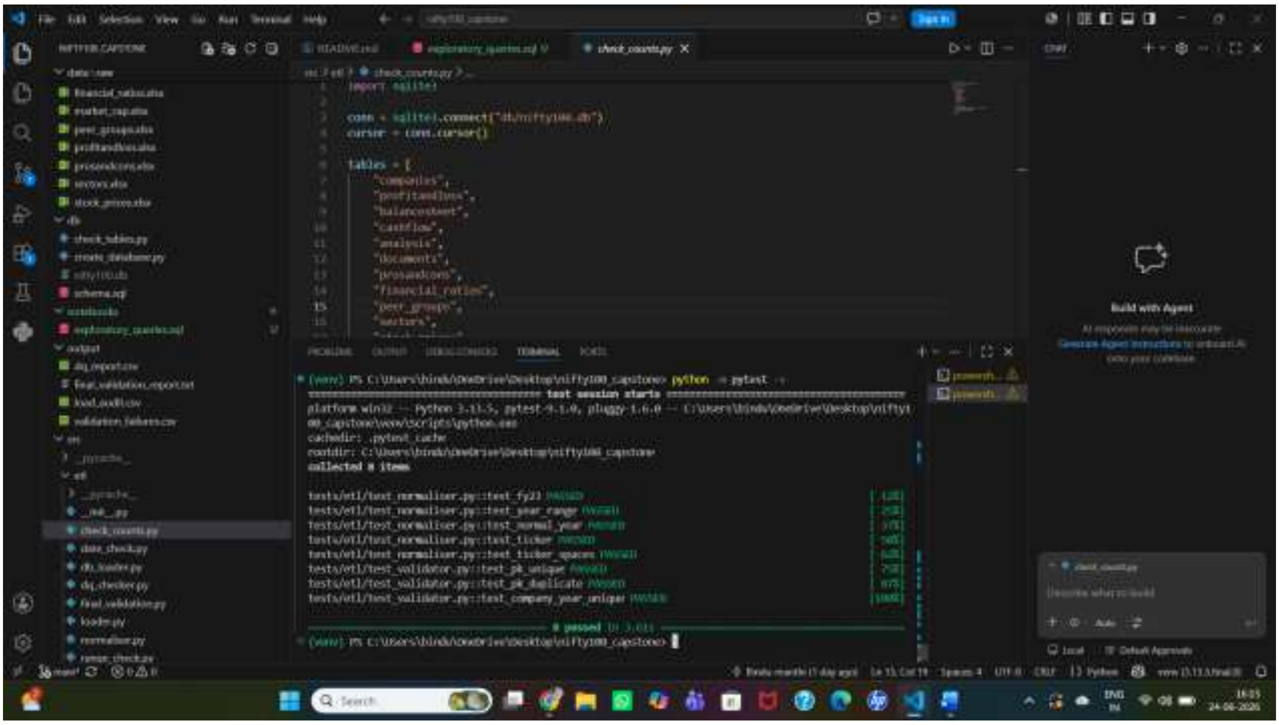

Figure 2: Successful execution of unit tests.

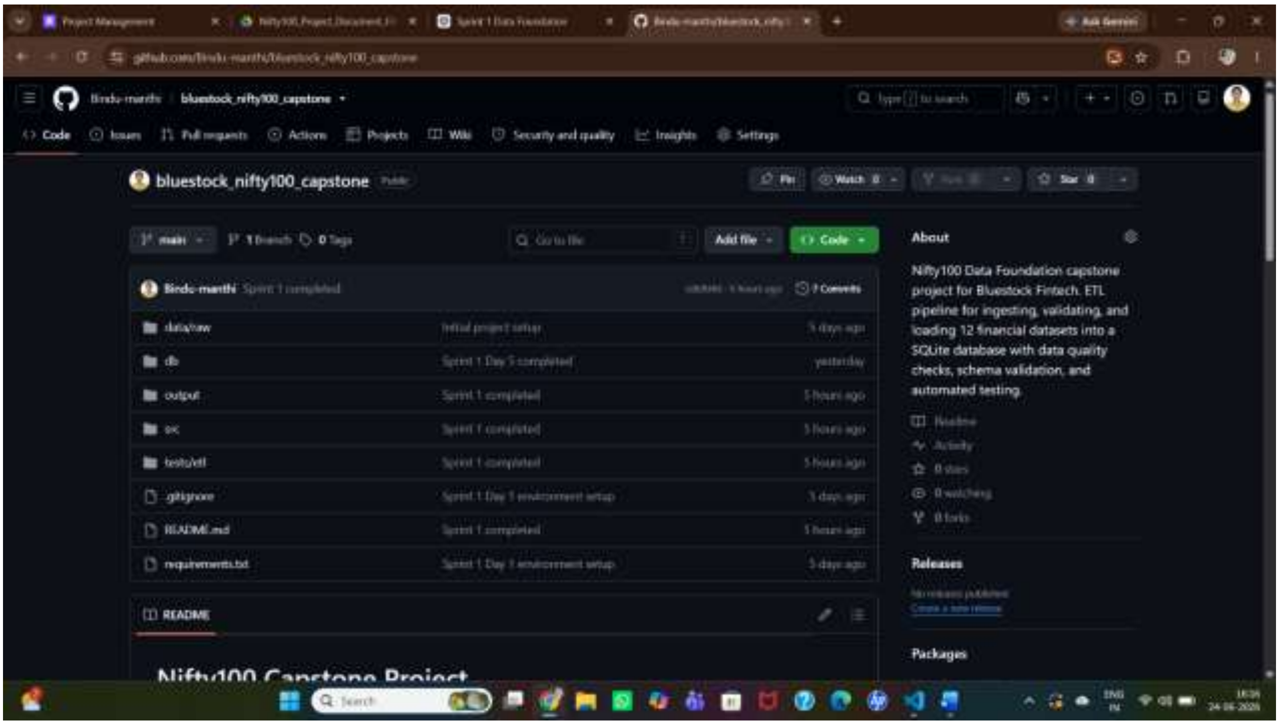

Figure 3: GitHub repository containing ETL scripts, database assets, validation reports, test cases, and project documentation for the Nifty100 Financial Intelligence Platform.

17.Conclusion

Sprint 1 successfully established the foundational data infrastructure for the Nifty100 Financial Intelligence Platform.

A robust ETL pipeline was developed, data quality checks were implemented, and a fully validated SQLite database was created. The project now provides a reliable foundation for future modules involving financial ratio analysis, stock screening, scoring systems, peer comparison, and dashboard reporting.

The objectives defined for Sprint 1 were achieved successfully, and the project is prepared for subsequent development phases.